

BRAC UNIVERSITY
CSE331 : Automata and Computability
Summer 2026

Duration: 25 minutes

Quiz 1

Total: 20 marks

Name:	ID:	0.5 Points
Section:		0.5 Points

Problem 1: State whether the following statement is True or False. If the statement is False, write the correct statement. (5 points)

- (1) In a DFA, the number of outgoing transitions from each state must be equal to $|Q|$, where Q is the set of states.
- (2) For a DFA M over alphabet Σ , the language $L(M)$ is the set of all strings over Σ that M accepts.
- (3) A language is regular if there exists a finite automaton that accepts at least one string from that language.
- (4) To prove that a DFA M recognizes a language L , it is enough to show that every string accepted by M is in L .
- (5) A string is accepted by a DFA if the computation visits an accept state at any point while reading the string.

1. **False.** Each state has exactly $|\Sigma|$ outgoing transitions, one for each input symbol.
2. **True.**
3. **False.** A language is regular if there exists a finite automaton that accepts **exactly all strings** in that language.
4. **False.** We must prove both:

$$w \in L(M) \Rightarrow w \in L$$

and

$$w \in L \Rightarrow w \in L(M).$$

5. **False.** A string is accepted only if the DFA finishes in an accept state after reading the entire string.

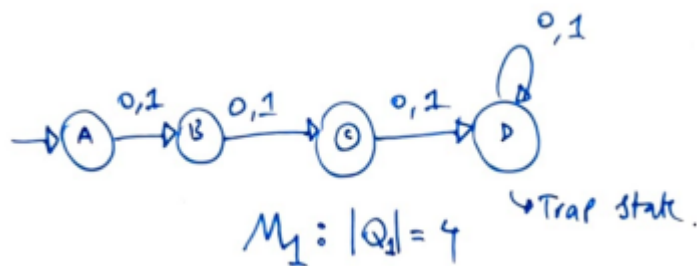
Problem 2: Designing DFAs (14 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

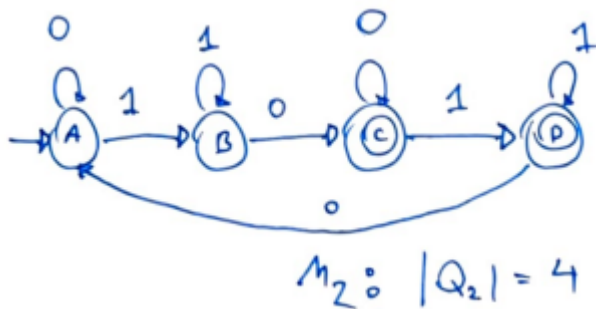
$L_1 = \{ w : \text{the length of } w \text{ is two} \}$

$L_2 = \{ w : \text{the number of times } 10 \text{ appears in } w \text{ is odd} \}$

(a) Give the state diagram for a DFA that recognizes L_1 . (3 points)



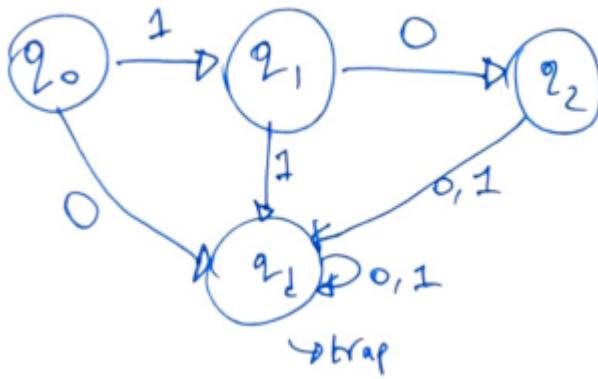
(b) Give the state diagram for a DFA that recognizes L_2 . (5 points)



(c) If we were to construct a DFA for the language $L_1 \cap L_2$ using the construction shown in class, how many states would it have? (1 point) $|Q_1| \times |Q_2| = 4 \times 4 = 16$

(d) How many strings are in $L_1 \cap L_2$? (1 point) ~~20~~ 1 only {10}

(e) Give a 4-state DFA for the language $L_1 \cap L_2$. (4 points)



Bonus: Closure under Union (2 points)

Prove that if A and B are regular languages, then $A \cup B$ is also regular.

Let $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognize A , and let $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognize B .

Using product construction, define a DFA $M = (Q, \Sigma, \delta, q_0, F)$ that recognizes $A \cup B$.

Specify Q , δ , q_0 , and F .

Let

$$M = (Q, \Sigma, \delta, q_0, F)$$

where

$$Q = Q_1 \times Q_2, \quad q_0 = (q_1, q_2),$$

$$\delta((r, s), a) = (\delta_1(r, a), \delta_2(s, a)),$$

and

$$F = (F_1 \times Q_2) \cup (Q_1 \times F_2).$$

Thus, M accepts when either M_1 or M_2 accepts. Therefore,

$$L(M) = A \cup B,$$

so $A \cup B$ is regular.